



Pinguecula: A Non-Communicable Diseases Eye Health Problem –

Causes, Risks, Prevention, and Management

Public Health Perspective on Conjunctival Degeneration

Submitted by

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1. Introduction and Problem Identification

1.1 Clinical Definition and Anatomical Characteristics

Pinguecula (plural: pingueculae) is a common, non-neoplastic degenerative condition of the bulbar conjunctiva — the transparent mucous membrane overlying the sclera (white of the eye). Clinically, it manifests as a small, yellowish, raised deposit of altered connective tissue, appearing preferentially on the nasal (medial) limbus and, less frequently, on the temporal (lateral) limbus within the interpalpebral zone. The lesion represents a focal accumulation of elastotic and hyalinized stromal collagen fibers beneath the conjunctival epithelium, often accompanied by calcium salt deposition in more advanced presentations (Arenas et al., 2020; Jaros & DeLuise, 1988).

Histopathologically, pinguecula is characterized by subepithelial elastotic degeneration of collagen — a process driven predominantly by cumulative actinic (ultraviolet) insult, biochemically distinct from normal aging. The condition is distinct from pterygium in that it remains confined to the conjunctival stroma and does not encroach upon the corneal epithelium or Bowman's layer. Nevertheless, pinguecula is widely recognized as a precursor lesion to pterygium, with nasal pingueculae carrying a proportionally higher risk of progression, attributable to differential intraocular UV reflection geometry (Arenas et al., 2020).

As illustrated in Figure 1, the characteristic clinical presentation involves bilateral, symmetrical yellowish deposits located at the 3 o'clock (temporal) and 9 o'clock (nasal) meridians of the conjunctival surface — the regions of greatest UV exposure due to their position within the interpalpebral fissure, the zone not shielded by the eyelid during normal gaze. This anatomical predilection is a defining diagnostic feature and underscores the etiological primacy of ultraviolet radiation in the pathogenesis of conjunctival degeneration.

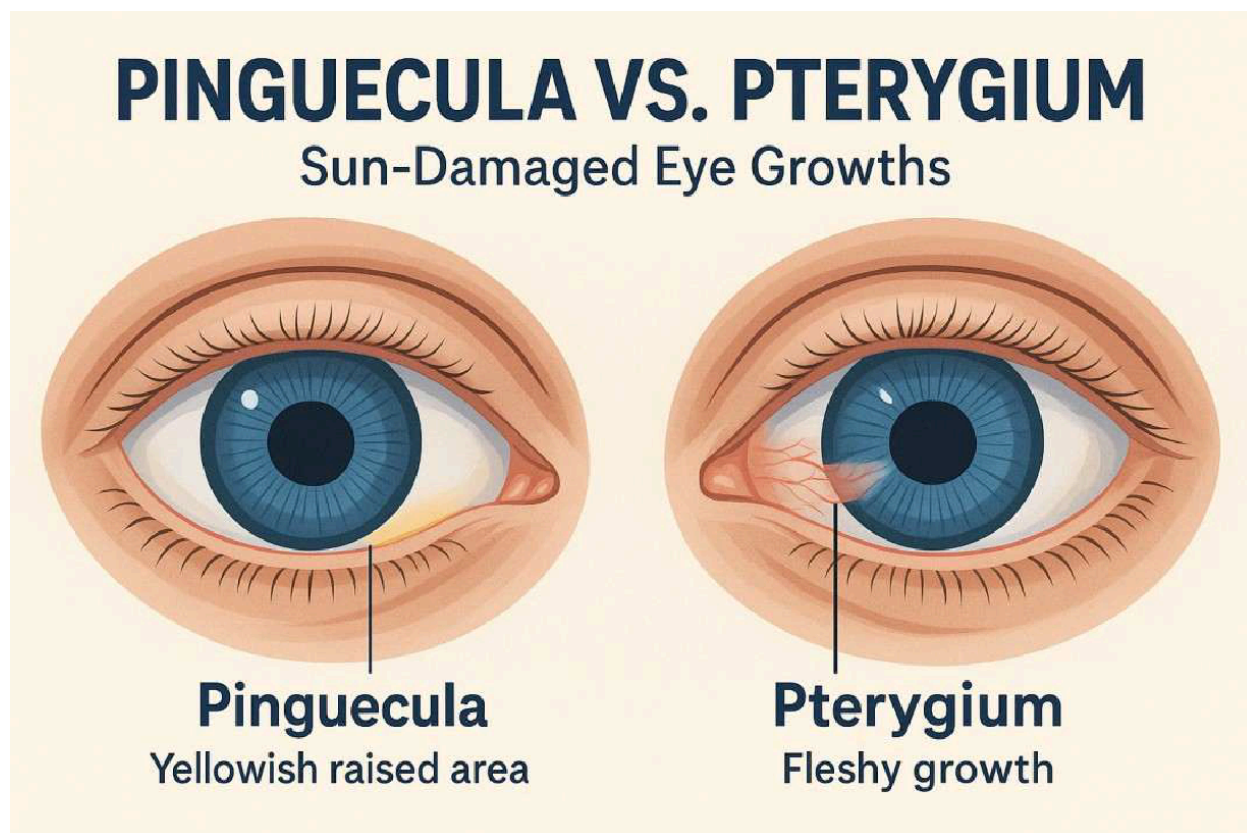


Figure 1. Bilateral pinguecula deposits at the nasal and temporal conjunctival margins within the interpalpebral fissure — the region of maximal UV exposure. (Source: Project presentation slides, Health for Life, 2026.)

1.2 Symptomatology and Clinical Presentation

Pinguecula follows a characteristically asymptomatic course in a substantial proportion of affected individuals, particularly in early-stage or small lesions. When symptoms do occur, they arise primarily from mechanical and inflammatory interactions between the elevated lesion and the overlying tear film. The principal symptom cluster encompasses: (i) ocular irritation and episodic redness (pingueculitis), driven by inflammation of surrounding conjunctival vasculature in response to environmental triggers such as wind, dust, or prolonged UV exposure; (ii) foreign body sensation — a persistent gritty or sandy ocular discomfort attributable to disruption of the preocular tear film over the raised lesion surface; (iii) dry eye syndrome, resulting from impaired tear film stability across the lesion; and (iv) epiphora (excessive tearing) as a reflexive lacrimal response to chronic ocular surface irritation (Sa-ngiampornpanit, 2025; Jaros & DeLuise, 1988).

In advanced cases or during acute inflammatory episodes, the lesion may adopt a hyperemic appearance and generate significant cosmetic concern. Progression to pterygium — a fibrovascular proliferative growth extending over the corneal surface — represents the most

clinically consequential endpoint of untreated pinguecula, with potential implications for visual acuity if the visual axis is involved. Nasal pingueculae carry a higher transition risk owing to UV focusing effects of the anterior ocular structures (Arenas et al., 2020).

1.3 Public Health Significance and Epidemiological Context

Despite its high global prevalence, pinguecula remains substantially underrecognized as a public health concern. In Thailand and other tropical nations, the condition is particularly prevalent owing to the country's geographical location between latitudes 5°N and 20°N — a region characterized by a persistently high annual UV index ($UVI > 10$), a large outdoor labor workforce, and limited widespread adoption of UV-protective eyewear. These conditions collectively constitute a substantial population-level burden of conjunctival degeneration that exceeds what clinical data alone may capture.

Population-based studies in analogous tropical and subtropical settings report pinguecula prevalence rates ranging from approximately 36% to 69.5% in adults over 49 years of age, with prevalence peaking in the sixth decade of life (60–69 years) — reflecting the cumulative, dose-dependent nature of UV-induced conjunctival damage (Panchapakesan et al., 1998). The condition disproportionately affects males, outdoor workers, rural residents, and individuals with lower educational attainment — population subgroups frequently underserved by preventive eye health programs. Even subclinical pinguecula carries meaningful quality-of-life impact through chronic low-grade ocular discomfort, reduced occupational productivity, and cumulative risk of progression to vision-threatening pterygium. The systematic under-reporting of this condition — partly because affected individuals normalize symptoms of dryness and irritation — further underscores the need for proactive public health surveillance and education, particularly in high-UVI tropical contexts such as Thailand.

2. Root Causes and Risk Factors

2.1 Pathophysiological Basis: Ultraviolet-Induced Conjunctival Degeneration

The primary etiopathogenic mechanism underlying pinguecula is chronic, cumulative exposure to ultraviolet (UV) radiation — specifically UVB (280–315 nm) and UVA (315–400 nm) wavebands — incident on the unprotected conjunctival surface. UV photons induce oxidative

stress within conjunctival stromal fibroblasts, generating reactive oxygen species (ROS) that damage collagen cross-linking and upregulate matrix metalloproteinases (MMPs). The resulting elastotic degeneration of the stroma — the replacement of normal fibrillar collagen with abnormal elastin-like material — constitutes the histological hallmark of the condition. In more advanced lesions, subepithelial calcium deposition may further harden and enlarge the growth (Arenas et al., 2020; Hatsusaka et al., 2021).

A critical concept in the epidemiology of pinguecula is the dose-response relationship between cumulative UV exposure and disease risk. Data from a longitudinal study of postmen in central Taiwan demonstrated that each incremental unit increase in lifetime cumulative ocular sunlight exposure is associated with a 2.10-fold increase in the odds of developing pinguecula (Tang et al., 1999, as cited in Hatsusaka et al., 2021). Analogous to UV-induced skin carcinogenesis, this dose-dependent relationship implies that the condition is fundamentally preventable through early and sustained reduction in cumulative UV exposure, and that protective behaviors initiated in early adulthood yield compounding long-term benefit.

2.2 Environmental Co-factors

While UV radiation constitutes the primary etiological driver, several environmental co-factors potentiate conjunctival degeneration through complementary mechanisms, often acting synergistically with UV exposure. Wind and particulate matter (dust, sand, agricultural debris) induce mechanical desiccation of the tear film and repetitive microtrauma to the conjunctival epithelium; industrial settings involving metal dust, cement, or chemical particulates further amplify this risk (Ay et al., 2022). Tobacco smoke and biomass combustion products introduce reactive chemical species directly onto the ocular surface, contributing to oxidative stromal damage. Low ambient humidity — prevalent in both arid and, paradoxically, air-conditioned tropical environments — reduces tear film stability, extending the duration of direct environmental exposure of the unprotected conjunctiva. An emerging body of evidence further suggests an association between ambient air pollution (particulate matter PM_{2.5}, ozone) and risk of pterygium and pinguecula-related hospitalization, indicating that chemical air pollutants may compound UV-driven ocular surface damage (Zeng et al., 2025).

2.3 Quantified Risk Factors: Epidemiological Evidence

A robust body of population-based epidemiological evidence from multiple geographic contexts — including Western Turkey, Taiwan, South India, Australia, and Eritrea — has identified and quantified the principal risk factors for pinguecula using odds ratio (OR) analysis. An OR greater than 1.0 indicates that the factor multiplies the probability of developing pinguecula relative to an unexposed reference group, while an OR less than 1.0 indicates a protective association. The key findings, presented in Table 1 and illustrated in Figure 2, are as follows.

Outdoor occupational work represents the single strongest quantified risk factor (OR = 5.00), reflecting the combined effect of prolonged daily UV exposure, environmental co-factors, and limited access to protective eyewear in occupational settings including agriculture, construction, fishing, and postal delivery (Dündar et al., 2020; Tesfai et al., 2021). Daily sun exposure exceeding two hours confers an OR of 4.03, an effect independent of occupational category, thereby implicating recreational and commuting UV exposure as independently significant behavioral risk factors. Rural geographic residence carries an OR of 2.94, consistent with greater outdoor activity duration, higher ambient UV in non-urban environments, and reduced access to protective eyewear services (Asokan et al., 2012). Male sex is associated with an OR of 1.50, a demographic disparity attributable to higher rates of outdoor occupational engagement and historically lower utilization of UV-protective eyewear among male populations.

Figure 2. Relative Risk of Pinguecula by Key Risk Factor (Odds Ratio)

Data sources: Western Turkey (Dündar et al., 2020), South India (Asokan et al., 2012), Taiwan (Tang et al., 1999)

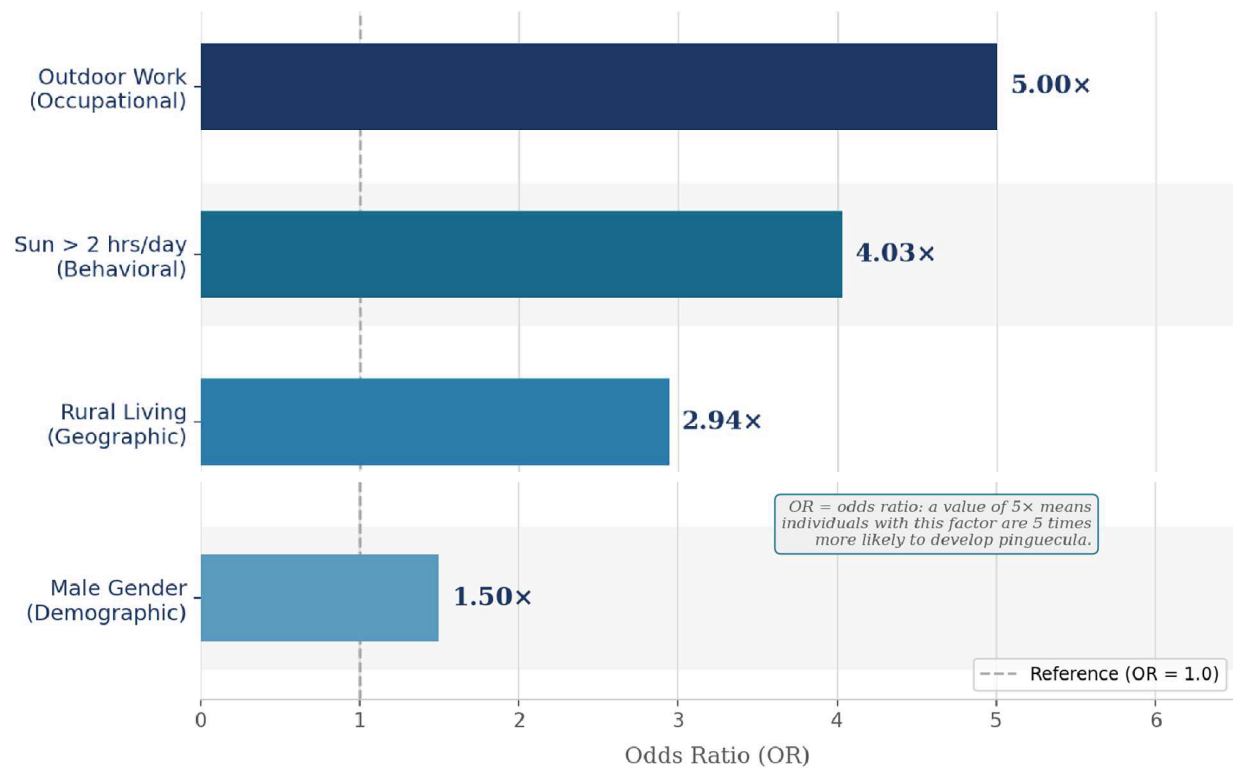


Figure 2. Relative risk of pinguecula expressed as odds ratios (OR) for four epidemiologically significant risk factors. Data derived from multi-country population-based studies (Dündar et al., 2020; Asokan et al., 2012; Tang et al., 1999).

Table 1. Quantified Risk and Protective Factors for Pinguecula: Multi-Study Odds Ratio Summary

Risk Factor	OR	Note	Source Region	Category
Outdoor occupational work	5.00×	Reference: indoor workers	W. Turkey / Australia	Occupational
Daily sun exposure > 2 hrs/day	4.03×	Significant at $p < 0.05$	Western Turkey	Behavioral
Rural geographic residence	2.94×	vs. urban reference	South India	Geographic
Each unit cumulative sunlight	2.10×	Dose-response per unit increase	Taiwan	Environmental
Male sex	1.50×	Consistent across cohorts	Multiple countries	Demographic

Risk Factor	OR	Note	Source Region	Category
Higher education (PROTECTIVE)	< 1.00	~43% risk reduction	Multiple countries	Protective
Indoor occupation (PROTECTIVE)	< 1.00	~80% risk reduction	Multiple countries	Protective

Note. OR = odds ratio. Values > 1.0 indicate increased risk; values < 1.0 indicate protective associations (green).
 Compiled from Dündar et al. (2020), Asokan et al. (2012), Tang et al. (1999), and Panchapakesan et al. (1998).

2.4 Age, Education, and Protective Demographic Factors

Age exhibits a non-linear relationship with pinguecula prevalence, reflecting the cumulative nature of UV damage: risk increases progressively with advancing age, peaking in the 60–69-year age group (Dündar et al., 2020). Prevalence may plateau or appear to decline beyond age 70 in some cohorts — a pattern attributable to survivor bias and differential mortality in highly exposed groups rather than biological risk reduction.

Educational attainment emerges as a significant protective factor in epidemiological models: individuals with higher education are approximately 43% less likely to develop pinguecula (OR < 1.00), likely mediated through greater health literacy, awareness of UV hazards, and higher adoption rates of protective eyewear (Asokan et al., 2012). Similarly, indoor occupational status is associated with an 80% reduction in pinguecula risk relative to outdoor work — reinforcing the primacy of UV exposure reduction as a prevention strategy. As depicted in Figure 3, these protective factors translate directly into evidence-based behavioral recommendations, including the adoption of 100% UV-blocking sunglasses, use of brimmed hats to attenuate superior indirect UV irradiance, and moisture control through avoidance of desiccating environments.

Figure 3. Evidence-Based Protective Factors Against Pinguecula

Factors with Odds Ratio (OR) < 1.0 indicate statistically significant reduction in pinguecula risk

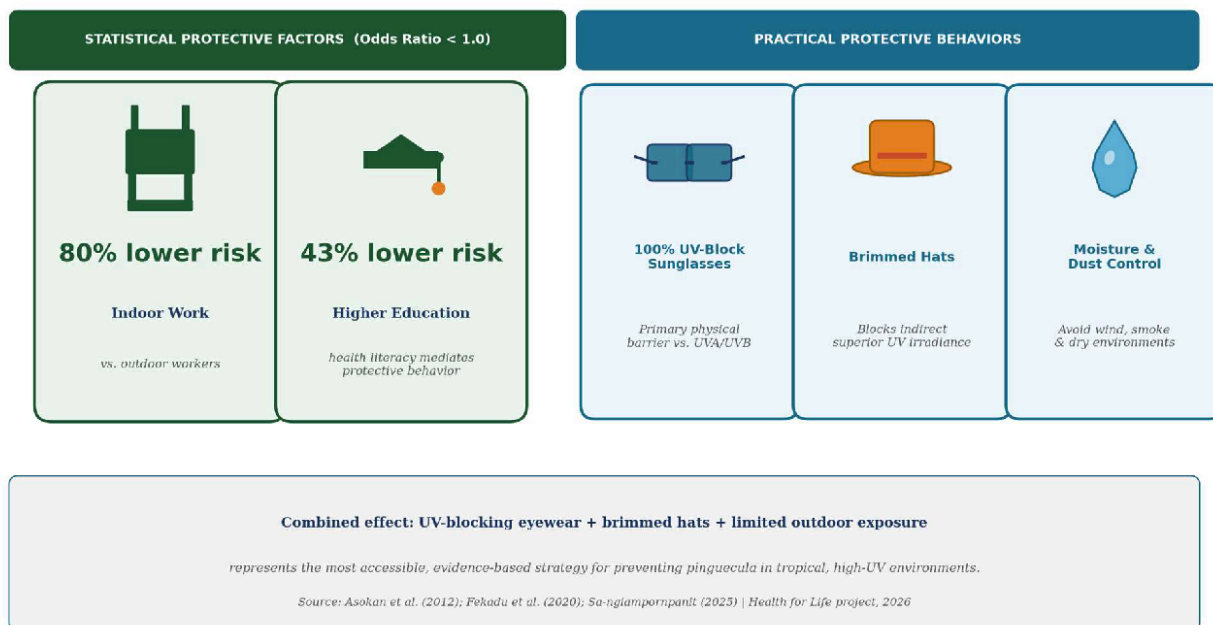


Figure 3. Evidence-based protective factors against pinguecula, stratified by statistical OR < 1.0 and practical behavioral interventions. (Source: Project presentation slides, Health for Life, 2026.)

3. Solutions and Mitigation Strategies

Management of pinguecula is stratified across three evidence-based tiers: primary prevention targeting exposure reduction before lesion formation, conservative medical management to relieve symptoms and control inflammation, and advanced or surgical intervention for refractory or cosmetically significant cases. This tiered approach reflects both the available evidence base and the cost-accessibility continuum relevant to a Thai public health context, as illustrated in Figure 4.

3.1 Primary Prevention: UV and Environmental Exposure Reduction

Given the dose-dependent relationship between cumulative UV exposure and pinguecula risk, primary prevention through ultraviolet radiation protection constitutes the most effective, evidence-based, and cost-efficient strategy available. Sunglasses providing 100% UV-A and UV-B blockage are the primary physical barrier against direct ocular UV radiation; epidemiological data confirm that their consistent use is associated with a statistically significant protective OR < 1.0 (Fekadu et al., 2020). Brimmed hats complement sunglass use by

attenuating indirect UV irradiance reaching the eye from superior angles — radiation that conventional sunglasses do not entirely intercept. Avoidance of prolonged outdoor exposure during peak UV hours (10:00–14:00), reduction of occupational sun exposure through indoor alternatives where feasible, and mitigation of environmental co-factors (wind, dust, tobacco smoke) collectively constitute first-line prevention (Sa-ngiampornpanit, 2025; Jaros & DeLuise, 1988).

From a public health perspective, primary prevention is particularly salient in Thailand, where a large proportion of the working-age population is employed in outdoor sectors (agriculture, construction, maritime industries) with high year-round UV burden. Integrating UV eye protection into occupational health regulations, and subsidizing UV-protective eyewear for outdoor workers, represents a scalable, low-cost policy intervention with potentially high population-level impact.

3.2 Conservative Medical Management: Symptom Control

For patients with established pinguecula experiencing symptomatic burden, conservative management aims to relieve discomfort and suppress episodic inflammation without modifying the lesion itself. Lubricating eye drops (artificial tears), viscous gels, and overnight ointments are the standard first-line pharmacological intervention, addressing tear film instability and the foreign body sensation arising from lesion-induced surface irregularity (Jaros & DeLuise, 1988). These preparations are available over-the-counter and may be used indefinitely without significant risk of systemic or ocular adverse effects.

During acute episodes of pingueculitis — characterized by pronounced conjunctival hyperemia, pain, and photophobia — short-course topical anti-inflammatory agents are indicated. Non-steroidal anti-inflammatory eye drops (NSAIDs) may be used for mild-to-moderate inflammation, while topical corticosteroids are reserved for more severe episodes, with careful monitoring for intraocular pressure elevation and cataract formation associated with extended steroid use. Intralesional corticosteroid injection (e.g., betamethasone) has been described in selected symptomatic cases, with case series reporting significant clinical improvement; however, the evidence base remains limited in terms of sample size and follow-up duration, precluding general recommendation at present (Arenas et al., 2020; Nejad et al., 2021).

3.3 Advanced and Surgical Interventions

Surgical intervention is rarely indicated for pinguecula in isolation, as the lesion itself does not threaten visual acuity. Operative management becomes appropriate when the lesion is (i) cosmetically unacceptable, (ii) associated with persistent symptoms refractory to conservative management, or (iii) progressing toward pterygium formation with imminent corneal risk. The current gold-standard operative procedure is surgical excision with conjunctival autograft, wherein the degenerated tissue is removed and the defect covered with healthy conjunctival tissue, typically secured with fibrin glue. This technique achieves favorable outcomes with lower recurrence than bare sclera excision (Nejad et al., 2021). In Thailand, this procedure is available at Bangkok Hospital at an estimated cost of approximately 45,000 THB, with lower-cost options at general hospitals (15,000–18,000 THB).

An emerging minimally invasive alternative is plasma-assisted noninvasive surgery (PANIS), utilizing an atmospheric low-temperature plasma device (Plexr) to sublimate pinguecula tissue without direct contact, incision, or sutures. A clinical case series of five eyes demonstrated complete lesion removal with full healing and no recurrence at six-month follow-up (Nejad et al., 2021). Key advantages include office-based delivery, elimination of operating room requirements, and potentially lower procedural costs. However, as of 2025–2026, PANIS is not offered as a standard service at Thai ophthalmic centers, and larger randomized controlled trials are required before it can be recommended as a routine treatment modality.

Figure 4. Cost-Tiered Management Pathway for Pinguecula

Ordered by cost and clinical complexity — from primary prevention through emerging surgical technology

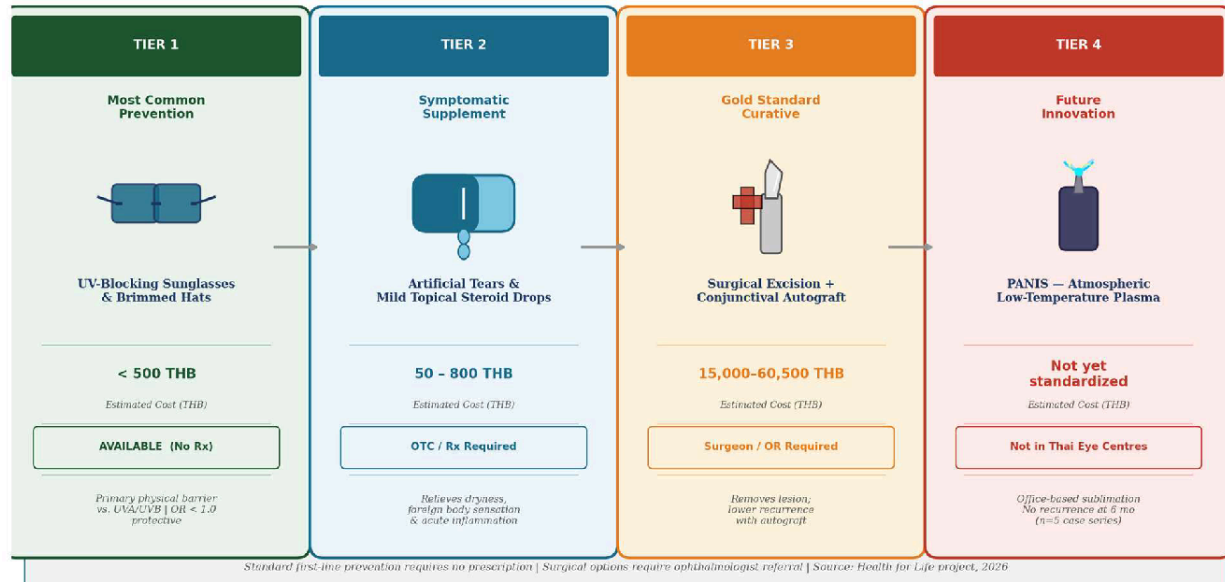


Figure 4. Cost-tiered management pathway for pinguecula, from primary prevention through emerging surgical technology. ✓ denotes approaches accessible without prescription. (Source: Project presentation slides, Health for Life, 2026.)

Table 2. Summary of Treatment and Management Approaches for Pinguecula

Treatment Approach	Primary Indication	Key Notes	Immediate (Non-Surg.)	Est. Cost (THB)
UV-blocking sunglasses & brimmed hats	Prevention / progression control	100% UV-block; also shields wind & dust	✓	< 500
Lubricating eye drops / artificial tears	Dryness, foreign body sensation	Standard first-line conservative therapy	✓	50–300
Topical steroids / NSAIDs	Acute pingueculitis (inflammation)	Short-term; risk of IOP elevation with steroids	✗	200–800
Intralesional corticosteroid injection	Symptomatic, unresponsive to topicals	Limited evidence; small case series only	✗	Varies
Surgical excision + conjunctival autograft	Severe/symptomatic; cosmetic concern	Gold standard; lower recurrence with autograft	✗	15,000–60,500

Treatment Approach	Primary Indication	Key Notes	Immediate (Non-Surg.)	Est. Cost (THB)
PANIS (atmospheric low-temperature plasma)	Emerging; complete lesion removal	Office-based; no recurrence at 6 mo (n=5); not yet standard in Thai centres	~	Not standardized

Note. ✓ = immediately accessible without prescription. ✕ = requires physician. ~ = emerging. THB = Thai Baht. Cost ranges: Bangkok Hospital, Bumrungrad, Rutnin Eye Hospital, Ladprao/Camillian Hospital (2025–2026).

3.4 Policy Landscape: The Absence of Pinguecula-Specific Public Health Programs

A critical contextual finding of this review is that, as of March 2026, there are no dedicated government policies or national campaigns in Thailand — nor in the broader international public health literature — that specifically target pinguecula or pterygium prevention as a named priority condition. Pinguecula is not currently recognized as a priority non-communicable disease in Thai Ministry of Public Health (MoPH) frameworks, and no formal pinguecula-specific drug pipeline or intervention program has been announced globally (Government Policy Review, 2026).

Notwithstanding this gap, several existing Thai governmental initiatives provide meaningful indirect mitigation of pinguecula risk through their UV exposure reduction components. These are summarized in Table 3. Most relevantly, the WHO SPECS 2030 initiative — launched by MoPH in November 2024 in partnership with WHO, the Thai Public Health Ophthalmology Society, and the Royal College of Ophthalmologists of Thailand — aims to improve nationwide access to quality eye care and integrates a public education pillar that supports UV-protection messaging applicable to degenerative conjunctival conditions (WHO, 2025). The Department of Disease Control's (DDC) annual heatstroke and UV warnings, issued each hot season (February–May), explicitly recommend wide-brimmed hats and UV-blocking sunglasses to outdoor populations, and directly reduce the lifetime UV exposure that constitutes the primary environmental trigger for pinguecula. The Ministry of Labour's Wet Bulb Globe Temperature (WBGT) occupational heat safety standard further indirectly benefits the outdoor worker population — precisely the demographic carrying a fivefold elevated pinguecula odds ratio.

It is notable that the most specific eye-targeted UV protection guidance in Thailand originates from the private sector — Bangkok Hospital, Rutnin Eye Hospital, and Samitivej Hospital have each published updated clinical articles explicitly recommending 100% UV-blocking sunglasses and hats for conjunctival degeneration prevention — rather than from formal government mandates. This asymmetry between private clinical guidance and public policy represents a structural gap that future national eye health programs, informed by the epidemiological evidence reviewed in this report, are well positioned to address.

Table 3. Relevant Thai Government Initiatives with Indirect Pinguecula Risk Reduction Benefit

Initiative / Program	Lead Agency	Relevance to Pinguecula	Status (2026)
WHO SPECS 2030 (launched Nov 2024)	Ministry of Public Health + WHO	Public eye-health education pillar supports UV-protection messaging (sunglasses, hats) covering degenerative conditions	Active rollout 2025–2026
DDC Annual Heatstroke & UV Warnings (Feb–May)	Dept. of Disease Control, MoPH	Explicitly recommends wide-brimmed hats and UV-blocking sunglasses; directly reduces cumulative UV exposure	Ongoing; 2026 issued March
Occupational Heat (WBGT) Safety Standard	Ministry of Labour	Limits sun exposure for outdoor workers (construction, agriculture) — the group with 5× pinguecula odds ratio	Enforced; revised 2016
Thai Meteorological Dept. Daily UV Index	Thai Meteorological Dept.	Daily UVI forecasts (routinely 11–12, Extreme); prompts general sun-avoidance advice across public channels	Ongoing public broadcast
Dept. of Health (Anamai) UV Infographics	Dept. of Health, MoPH	"6 Ways to Protect Yourself from Dangerous UV Rays" (updated 2024); recommends avoiding peak sun, umbrellas, protective clothing	Updated 2024; recirculated

Note. No Thai government program currently names pinguecula or pterygium as a targeted condition. All initiatives provide indirect benefit through broader UV exposure reduction. Sources: WHO (2025); DDC, MoPH (2026); Ministry of Labour (2016); Thai Meteorological Department; Department of Health, MoPH (2024).

4. Case Studies

The following two case studies document real-world programmatic responses to UV-related ocular surface disease — conditions directly analogous to and often comorbid with pinguecula. As pinguecula-specific public health interventions remain sparse in the published literature, these cases address pterygium and UV-related eye degeneration more broadly; the shared etiological pathway (chronic UV exposure) renders these programs directly applicable to the prevention and burden reduction of pinguecula.

Case Study 1: Australia's SunSmart Campaign — National Sun and UV Protection Program

Project Overview

Objectives

The SunSmart campaign, launched in 1981 by the Cancer Council Victoria and subsequently adopted nationally across Australia, was designed to reduce population-level UV-related skin and eye damage — including pterygium, cataracts, and pinguecula — through sustained public awareness and measurable behavior change. It represents one of the longest-running and most rigorously evaluated UV protection programs globally.

Key Activities and Implementation

The campaign employed a multi-channel mass media strategy anchored by the mnemonic 'Slip, Slop, Slap, Slide, Seek' (covering protective clothing, sunscreen, hats, sunglasses, and shade-seeking behavior respectively). Interventions were delivered across school curricula, occupational health programs in high-risk industries (agriculture, construction, lifeguarding), community events, and national broadcasting. School-based components introduced UV eye protection education from primary age, embedding protective behavior in early developmental stages. Occupational programs specifically promoted wrap-around UV-blocking sunglasses and wide-brimmed hats for outdoor workers — the highest-risk demographic for conjunctival UV damage.

Results and Outcomes

Epidemiological analyses of pterygium surgery rates in Australia demonstrated an approximately 47% national decline between 1994 and 2017, a trend temporally correlated with the expansion and sustained delivery of the SunSmart program. Population surveys documented significant increases in sunglasses and hat usage in outdoor settings among adults and children. The program provides a replicable public health model for tropical countries such as Thailand, demonstrating that sustained, multi-sector UV protection campaigns can achieve measurable reductions in the burden of UV-related ocular surface disease at a population level.

Case Study 2: Misión Milagro — Cuba–Venezuela Ophthalmic Surgical Outreach Initiative

Project Overview

Objectives

Misión Milagro ('Operation Miracle') was a bilateral humanitarian initiative between Cuba and Venezuela, launched in 2004, with the primary objective of providing free ophthalmic surgical care — including pterygium and pinguecula excision — to low-income patients with vision-impairing ocular conditions in resource-limited communities across Venezuela and subsequently across Latin America and the Caribbean. The initiative directly addressed the systemic access barriers that disproportionately prevent low-income and rural populations from receiving care for UV-related ocular conditions.

Key Activities and Implementation

The program deployed mobile and fixed-site surgical teams staffed by Cuban ophthalmologists, conducting community-based screening to identify candidates for pterygium excision, cataract removal, and other procedures. Community health workers facilitated patient identification, transportation, and postoperative follow-up. The program operated on a zero-cost model for patients, directly addressing the financial barrier — analogous to the 15,000–60,000 THB cost of pinguecula excision in Thailand — that prevents many affected individuals in low- and middle-income settings from accessing surgical care.

Results and Outcomes

Reported outcomes from Venezuelan intervention sites indicated that approximately 90.9% of pterygium surgical cases achieved clinically significant improvement in visual acuity and ocular surface health. The program demonstrated that high-volume, high-quality ophthalmic surgery for UV-related ocular conditions is feasible in resource-poor settings when financial and geographic barriers are systematically removed. For contexts such as rural Thailand — where outdoor occupational populations face combined high UV exposure risk and limited affordable surgical eye care — this model offers a compelling template for government or NGO-funded ophthalmic outreach programming.

5. Conclusion and Recommendations

5.1 Summary of Key Findings

Pinguecula is a common, largely preventable degenerative condition of the conjunctiva driven primarily by cumulative UV radiation exposure, with documented compounding risk from occupational, geographic, behavioral, and demographic factors. Epidemiological evidence from multi-country studies consistently demonstrates that outdoor workers, rural residents, older adults, and males are at substantially elevated risk, with odds ratios reaching up to 5× for outdoor occupational exposure. The condition is currently underrecognized in public health discourse despite its significant quality-of-life impact and its role as a precursor to the vision-threatening pterygium. Treatment options range from accessible, low-cost preventive measures through

conservative pharmacological management to surgical excision, with an emerging plasma-based minimally invasive technique offering promise for future clinical practice.

A critical finding of this review is the complete absence of pinguecula-specific public health policies at both the Thai national level and globally. Despite Thailand's high-UVI tropical environment and its large outdoor workforce, no dedicated prevention programs exist. The indirect mitigation provided by existing UV/heat advisories and the WHO SPECS 2030 initiative, while valuable, falls short of the targeted, sustained intervention that the evidence base supports. International examples, particularly Australia's SunSmart campaign, demonstrate that such targeted programs are feasible and produce measurable population-level outcomes.

5.2 Recommendations for Public Health Action

Based on the synthesized evidence, the following recommendations are proposed for public health policy and practice, with particular relevance to the Thai tropical context:

- **Establish pinguecula as a recognized NCD sub-category in Thai national eye health policy:** Recognition within the MoPH non-communicable disease framework would enable dedicated surveillance, prevention guidelines, and resource allocation.
- **Integrate UV eye protection into national and occupational health campaigns:** Thailand's existing public health infrastructure should explicitly incorporate 100% UV-blocking sunglasses and brimmed hat recommendations with specific reference to conjunctival degeneration risk.
- **Subsidize UV-protective eyewear for high-risk outdoor workers:** Given outdoor occupational OR = 5.0 and the low cost of UV-blocking sunglasses, employer- or government-subsidized provision for outdoor laborers constitutes a high-return public health investment.
- **Strengthen school and community eye health education:** The 43% risk reduction associated with higher educational attainment supports integration of age-appropriate ocular UV protection education into school curricula, establishing protective habits from early life.

- **Establish systematic monitoring of pinguecula-to-ptyerygium progression:** Routine ophthalmic screening for high-risk occupational groups should incorporate standardized pinguecula grading protocols, enabling earlier intervention before vision-threatening pterygium develops.
- **Invest in research on emerging treatments:** Larger randomized controlled trials of PANIS with extended follow-up are required. Thai ophthalmic centers are well positioned to contribute given high local prevalence of conjunctival degeneration.

In summary, pinguecula exemplifies a condition in which the public health burden substantially exceeds current clinical recognition and policy response. Simple, low-cost preventive measures — most fundamentally, consistent use of UV-blocking sunglasses and hats — are capable of preventing the majority of new cases if adopted at scale. Translating this evidence into structured national campaigns, occupational health policy, and equitable access to care represents the most actionable near-term opportunity to reduce the burden of UV-related conjunctival disease in Thailand and comparable tropical settings.

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Appendices

The following appendices supplement the main report with consolidated statistical data, comprehensive treatment and policy reference tables, and a figure source index, as required by the Health for Life course assessment criteria (Project Details, 2026).

Appendix A1. Consolidated Odds Ratio Data

Table A1 consolidates the epidemiological odds ratio data presented in Section 2. An OR > 1.0 denotes a risk-increasing association; an OR < 1.0 denotes a protective association. All values are derived from multi-country population-based studies as cited in the References section.

Table A1. Risk and Protective Factor Odds Ratios for Pinguecula Development

Risk Factor	OR	Note	Source Region	Category
Outdoor occupational work	5.00×	Reference: indoor workers	W. Turkey / Australia	Occupational
Daily sun exposure > 2 hrs/day	4.03×	Significant at $p < 0.05$	Western Turkey	Behavioral
Rural geographic residence	2.94×	vs. urban reference	South India	Geographic
Each unit cumulative sunlight	2.10×	Dose-response per unit increase	Taiwan	Environmental
Male sex	1.50×	Consistent across cohorts	Multiple countries	Demographic
Higher education (PROTECTIVE)	< 1.00	~43% risk reduction	Multiple countries	Protective
Indoor occupation (PROTECTIVE)	< 1.00	~80% risk reduction	Multiple countries	Protective

Note. OR = odds ratio. PROTECTIVE factors highlighted in green (OR < 1.0). Data: Dündar et al. (2020); Asokan et al. (2012); Tang et al. (1999); Panchapakesan et al. (1998).

Appendix A2. Comprehensive Treatment Reference Table

Table A2 provides a full reference of current and emerging treatment approaches for pinguecula, organized by clinical indication, immediacy of non-surgical availability, and estimated cost in Thai Baht. Approaches marked ✓ are accessible without clinical prescription and represent the primary prevention and self-management tier.

Table A2. Comprehensive Treatment and Management Options for Pinguecula

Treatment Approach	Primary Indication	Key Notes	Immediate (Non-Surg.)	Est. Cost (THB)
UV-blocking sunglasses & brimmed hats	Prevention / progression control	100% UV-block; also shields wind & dust	✓	< 500
Lubricating eye drops / artificial tears	Dryness, foreign body sensation	Standard first-line conservative therapy	✓	50–300
Topical steroids / NSAIDs	Acute pingueculitis (inflammation)	Short-term; risk of IOP elevation with steroids	✗	200–800
Intralesional corticosteroid injection	Symptomatic, unresponsive to topicals	Limited evidence; small case series only	✗	Varies
Surgical excision + conjunctival autograft	Severe/symptomatic; cosmetic concern	Gold standard; lower recurrence with autograft	✗	15,000–60,500
PANIS (atmospheric low-temperature plasma)	Emerging; complete lesion removal	Office-based; no recurrence at 6 mo (n=5); not yet standard in Thai centres	~	Not standardized

Note. ✓ = immediately accessible without prescription. ✗ = requires physician. ~ = emerging. THB = Thai Baht. Cost ranges: Bangkok Hospital, Bumrungrad, Rutnin Eye Hospital, Ladprao/Camillian Hospital (2025–2026 estimates).

Appendix A3. Thai Government Policy Reference Table

Table A3 provides a consolidated reference of all identified Thai governmental initiatives with relevance to pinguecula risk reduction as of March 2026. No program explicitly names pinguecula as a target condition; all listed initiatives confer indirect benefit through broader UV and eye-health frameworks.

Table A3. Thai Government Initiatives with Indirect Pinguecula Risk Reduction Benefit

Initiative / Program	Lead Agency	Relevance to Pinguecula	Status (2026)
WHO SPECS 2030 (launched Nov 2024)	Ministry of Public Health + WHO	Public eye-health education pillar supports UV-protection messaging	Active rollout 2025–2026

Initiative / Program	Lead Agency	Relevance to Pinguecula	Status (2026)
		(sunglasses, hats) covering degenerative conditions	
DDC Annual Heatstroke & UV Warnings (Feb–May)	Dept. of Disease Control, MoPH	Explicitly recommends wide-brimmed hats and UV-blocking sunglasses; directly reduces cumulative UV exposure	Ongoing; 2026 issued March
Occupational Heat (WBGT) Safety Standard	Ministry of Labour	Limits sun exposure for outdoor workers (construction, agriculture) — the group with 5× pinguecula odds ratio	Enforced; revised 2016
Thai Meteorological Dept. Daily UV Index	Thai Meteorological Dept.	Daily UVI forecasts (routinely 11–12, Extreme); prompts general sun-avoidance advice across public channels	Ongoing public broadcast
Dept. of Health (Anamai) UV Infographics	Dept. of Health, MoPH	"6 Ways to Protect Yourself from Dangerous UV Rays" (updated 2024); recommends avoiding peak sun, umbrellas, protective clothing	Updated 2024; recirculated

Note. Sources: WHO (2025); DDC, MoPH (2026); Ministry of Labour (2016); Thai Meteorological Department; Department of Health, MoPH (2024).

Appendix A4. Figure and Slide Source Index

All figures referenced throughout this report are sourced from the group's presentation slides submitted for the Health for Life course (March 2026).

Figure	Description	Section	Slide Reference
Figure 1	Anatomical eye diagram — bilateral pinguecula at nasal/temporal meridians	Section 1.1	Slide: '3 Things about Pinguecula'
Figure 2	Odds ratio bar chart — four primary risk factors	Section 2.3 / App. A1	Slide: 'Statistical Risk Factors: UV & Occupation'
Figure 3	Protective factors infographic — OR < 1.0 and behavioral interventions	Section 2.4	Slide: 'Reducing the Odds: Evidence-Based Prevention'

Figure	Description	Section	Slide Reference
Figure 4	Cost-tiered treatment pathway — four management levels	Section 3.3	Slide: 'Preventing the problem can begin now'

*Note. Placeholder brackets in the report body indicate insertion points for the corresponding slide images.
(Apichaikriengkrai, Sirithienthong & Phonglaungtham, 2026.)*